

Vinh Ly

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Summary

Staff software engineer with **14 years of experience** building and maintaining large-scale backend systems and consumer-facing web applications. Focused on **Python** as the primary language, with significant work in **Node.js** and **React** frontend development. Spent the last decade at Google and YouTube, where reliability, scale, and clean API design were the main constraints. Comfortable working across the full stack when the problem calls for it.

Core Skills

Languages: Python, Node.js, TypeScript, Go, Java

Frontend: React, Vue.js, HTML/CSS, SASS, WebSockets

Backend & APIs: REST APIs, GraphQL, OAuth2, JWT, gRPC

Data: MySQL, PostgreSQL, Redis, MongoDB, BigQuery

Infra & Tooling: Docker, Kubernetes, GCP, AWS, CI/CD pipelines, Git

Work Experience

Staff Software Engineer | YouTube | Remote | 02/2025 - Present

- Working on backend services powering video ingestion and content delivery pipelines. Most of the work involves **Python** microservices handling millions of requests per minute.
- Built and maintained internal tooling for monitoring pipeline health and automating incident response workflows.

Staff Software Engineer | Whatnot | Remote | 05/2024 - 02/2025

- Joined the marketplace platform team to help scale the live commerce infrastructure during a period of rapid growth. Worked on **Python** services handling checkout, inventory, and real-time bid processing.
- Refactored core API endpoints that were causing latency spikes during high-traffic events. Introduced Redis caching and query optimization; **p99 latency dropped from over 2 seconds to under 300ms** for the most frequently called endpoints.

Staff Software Engineer / Manager | Google | Remote | 03/2015 - 04/2024

- Led a team of 5 engineers building and maintaining internal developer productivity tools used by thousands of Googlers daily. The main codebase was in **Python**, with a web UI in **React**.
- Migrated a critical internal service from Python 2 to Python 3, which required coordinating with 12 downstream teams that depended on the API. The migration finished on schedule with 0 user-facing incidents.
- Owned the reliability and performance of the service for several years. **Uptime stayed above 99.95%** across all of 2023. When incidents did happen, the on-call runbooks I wrote cut mean time to resolution by roughly 40%.
- Served as technical lead for a cross-functional project involving the DevOps, security, and SRE teams to harden the service against a new class of internal compliance requirements. Designed the authentication layer using OAuth2 and introduced structured audit logging.
- Conducted technical interviews and mentored three junior engineers, two of whom were promoted to mid-level within 18 months.

Software Engineer | Google | Remote | 02/2011 - 02/2015

- Worked on backend infrastructure for an internal content management system. Built new features in **Python** and maintained a **Node.js** service that handled real-time notifications for the UI.

- Improved the CI/CD pipeline for the team, cutting average build-and-test time from around 25 minutes to under 8 minutes by parallelizing test suites and introducing incremental builds.
- Collaborated closely with product managers and UX researchers to ship three major UI redesigns of the internal tool. The frontend was rebuilt in **React** during one of those cycles.

Software Design Engineer II | Microsoft | Remote | 10/2007 - 12/2011

- Worked on developer tools for the Visual Studio platform. Primarily wrote **Python** scripts for build automation and testing infrastructure, with some C# for the IDE extensions themselves.
- Reduced flaky test failures in the CI system by redesigning the test isolation strategy. **Flaky test rate fell from around 8% to under 1%** over the course of a quarter.

Software Design Engineer Intern | Microsoft | Remote | 06/2006 - 08/2006

- Summer internship on the SQL Server tools team. Built a reporting plugin for the management studio in C#.

Education

BS Computer Science | University of Texas at Austin | 2003 - 2007